

Week 5 Task

Introduction to Wireless Communication

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Code Summary:

We designed and simulated a custom wireless Reinforcement Learning (RL) problem using the Gymnasium framework. The **Agent** acts as a transmitter whose objective is to adjust its transmit power level to hit a target **SNR of 12 dB** at the receiver, while simultaneously minimizing unnecessary power consumption.

- **Environment:** Models a wireless channel influenced by random path loss (fading) and static background thermal noise.
- **Action Space:** Discrete with 5 possible transmit power levels: [10, 14, 18, 22, 26] dBm.
- **State Space:** Consists of 11 discrete states representing the receiver's measured SNR. Continuous float SNR values from the channel are mapped to the nearest pre-defined bin center ranging from 0 to 20 dB (with a step size of 2 dB).
- **Algorithmic Approach:** We implemented tabular **Q-Learning** trained over 1,000 episodes. Action selection utilizes an ϵ -greedy policy configured with a 10% exploration rate (to discover channel characteristics) and a 90% exploitation rate.

The output of the training process yields an optimized 11×5 **Q-Table** matrix, representing the learned policy that the transmitter uses to map any incoming SNR state to the most power-efficient transmit action.

Code Output:

```
Training finished! Final optimized Q-Table:
[[ -8.75135617  -8.12714622  -7.88053521  -7.96590563  -8.42961289]
 [ -6.39232895  -6.47917472  -7.28694864  -6.32281698  -6.89203033]
 [ -9.36164824  -8.66992356  -8.76993781  -8.46954524  -8.5852771 ]
 [ -8.09601403  -8.19037421  -8.09233148  -7.71594157  -7.78207857]
 [ -9.73457366  -9.65608507  -9.87491921  -9.7239882  -9.71504877]
 [ -9.88951927  -9.90405786  -9.46729984  -9.39350105  -9.58454072]
 [-10.78319798 -10.01497442 -10.14716991  -9.96137744  -9.78309582]
 [ -9.54273605  -9.83064297  -8.90367932  -9.23962394  -9.01102685]
 [ -9.68653513  -9.95992091  -9.76169826  -9.87557457  -9.43275935]
 [ -6.03241034  -6.93012403  -5.79178395  -6.18326249  -5.82665988]
 [ -7.09527599  -6.70496672  -6.18961705  -6.25798317  -6.11758242]]
```